

UNiSO: Privacy-First Social Discovery Engine

1. Project Overview: UNiSO (Research Edition)

UNiSO is a decentralized social networking platform designed for the Gen Z era. Unlike traditional social media that harvests user data on central servers, UNiSO utilizes Federated Learning to provide hyper-personalized networking while keeping sensitive student data strictly on-device.

2. The Concept: Decentralized Intelligence

1. Local Training: The application uses the mobile's Neural Processing Unit (NPU) to train a personalization model based on local interaction and performance data.
2. Privacy-First Aggregation: Only mathematical 'model weights' are sent to the server. No raw data, chats, or grades ever leave the phone.
3. Global Optimization: A central coordinator merges these weights to improve the matching algorithm for everyone without ever seeing individual profiles.

3. Research Influence (Dr. M S Bhargavi)

- FEdRank (2025): Collaborative Institutional Ranking using Borda Count to rank student performance metrics without violating privacy.
- FedWAPR (2025): Weighted Aggregation via Probability-based Ranking to ensure only high-quality, non-malicious updates affect the global model.

4. Technical Architecture

- Frontend: React Native with TensorFlow.js (LiteRT).
- Backend: Node.js (API) + Python/FastAPI (Federated Aggregator).
- FL Framework: Flower (flwr.dev) for cross-platform model coordination.

5. Reference Links

[Dr. Bhargavi Research: FedWAPR \(Elsevier 2025\)](#)

[Dr. Bhargavi Research: FEdRank \(IEEE 2025\)](#)

[Federated Learning Framework: Flower](#)

[Mobile AI: TensorFlow Lite/LiteRT](#)